

Report R5 — Dissipative rules: recurrent classes, attractors, and protected coherence

Krylov sector analysis of quantum cellular automata (Tier 1a/1b)

QCA Sector Analysis project (Tier 1)

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Abstract

We extend the exact pipeline to the 240 dissipative quantum cellular automata (rules containing a reset channel D or E). The one-cycle channel is now genuinely non-unitary: the directed transition graph has *transient* strongly connected components flowing into *recurrent classes* (attractors). We sweep all dissipative rules (both boundary conventions, $N = 6 \dots 13$), census their attractor structure, and classify each recurrent class by the peripheral spectrum of its exact restricted superoperator (Tier 1b) as a classical pointer state (*mixing*), a *limit cycle*, or a *coherence-carrying* attractor. Finally we quantify, exactly at small N , the *protected-coherence gap* — the amount by which the full channel’s fixed-point space exceeds the classical pointer count — which the transition graph provably under-resolves for weakly dissipative rules.

1 Dissipative graph structure

For a dissipative rule the channel $\Phi_C(\rho) = \sum_{\mu} K_{\mu} \rho K_{\mu}^{\dagger}$ has ≥ 2 Kraus operators. The exact engine propagates every Kraus branch (never merging incoherent alternatives) and records the directed transition graph $M_{yx} = \sum_{\mu} |\langle y | K_{\mu} | x \rangle|^2$. Its terminal strongly connected components are the recurrent classes (attractors); non-terminal components are transient. We record, per (rule, N , bc): the number and sizes of recurrent classes, the transient depth (longest condensation-DAG path), and the basin structure (states reaching a single vs. several attractors). The engine and its dissipative branch machinery are validated against an independent dense-channel oracle (528/528 support checks; per-site Kraus completeness $A_0^{\dagger} A_0 + A_1^{\dagger} A_1 = \mathbb{K}$).

2 Census of attractor structure

Table 1 classifies every dissipative rule by the structure at the largest computed N : a unique classical fixed point, several point attractors (multistability, signalling a strong basis-diagonal conserved charge), or one or more extended (> 1 -state) attractors.

attractor structure	obc0	pb
unique fixed point	66	36
multistable (point attractors)	77	75
single extended attractor	63	47
multiple extended attractors	34	82
total dissipative rules	240	240

Table 1: Dissipative rules by attractor structure and boundary convention.

Two ground-truth items (context §8) are reproduced: rule 22 (IVVD) on **pb**c at even N has exactly three point attractors (the vacuum and the two Néel states); rules 28/29 illustrate the boundary/wall-transparency contrast.

3 Tier 1b: attractor types

For each recurrent class R we build the *exact* restricted superoperator Φ_R on $\text{span}(R) \otimes \text{span}(R)^*$ from the labelled Kraus branches, and read its peripheral spectrum (geometric multiplicities, robust to the non-normal transient Jordan structure). A class is *mixing* (a unique steady state; a classical pointer/dark state when $|R| = 1$), a *limit cycle* (peripheral roots of unity, basis-diagonal eigenoperators), or *coherence-carrying* (a decoherence-free subsystem of dimension $d_k > 1$). Table 2 surveys representative rules.

W	tuple	#recur	attractor kinds	max d_k
0	DDDD	1	1×mixing	1
1	VDDD	1	1×unresolved	1
22	IVVD	3	3×mixing	1
28	IIVD	11	3×mixing, 8×coherent	3
29	EIVD	10	2×mixing, 8×coherent	3
50	DVVV	1	1×mixing	1
76	IIID	131	131×mixing	1
178	DVVE	2	2×mixing	1
200	DIII	90	90×mixing	1
232	DIIE	46	46×mixing	1

Table 2: Tier-1b attractor classification (pb, $N = 8$). Rule 28 carries coherence-protecting extended attractors ($d_k > 1$); most others relax to classical pointer states.

4 Protected coherence: the Cesaro gap

The transition graph sees only *strong* (basis-diagonal) conserved charges; *weak* symmetries protect coherences that the graph misses (context §5). We quantify this exactly at small N by the *Cesaro rank* — the geometric multiplicity of eigenvalue 1 of the full 4^N -dimensional channel superoperator, i.e. the dimension of the steady-state space — and compare it to the graph-resolved dimension $\sum_R \dim \text{Fix}(\Phi_R)$ obtained by summing the per-recurrent-class fixed-point dimensions (Table 3). The difference is the extra cross-class / weak-symmetry protected coherence, which is non-negative and invisible to the transition graph.

W	tuple	classical dim	Cesaro rank	coherence gap
0	DDDD	1	1	0
1	VDDD	15	15	0
22	IVVD	3	25	22
28	IIVD	3	9	6
50	DVVV	1	13	12
178	DVVE	2	9	7
200	DIII	10	100	90
232	DIIE	6	36	30

Table 3: Exact Cesaro rank vs. classical pointer dimension (pb, $N = 4$). A large gap (rule 22: 25 vs. 3; rule 50: 13 vs. 1) flags a weakly dissipative rule whose channel protects extensive coherence invisible to the graph; a zero gap (rule 0, pure decay) is fully classical.

5 Transient relaxation

Figure 1 shows the transient depth (an exact integer proxy for the relaxation time) growing with N for representative multistable rules.

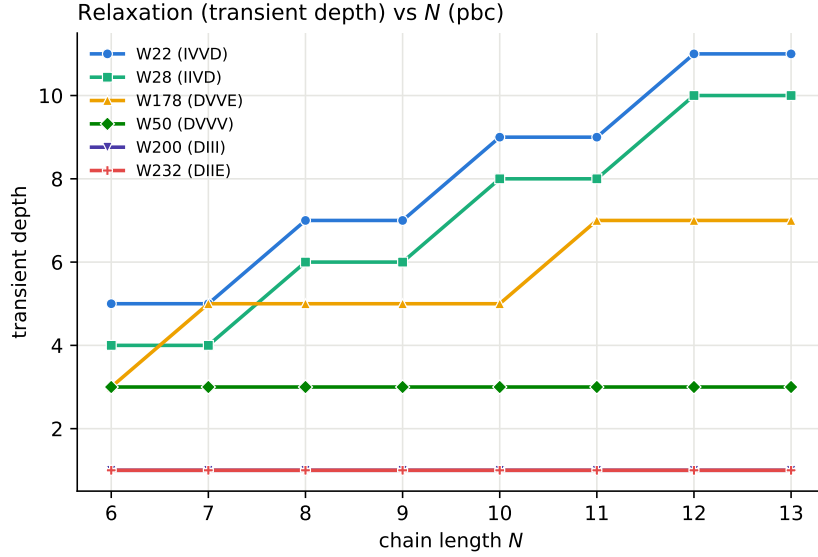


Figure 1: Transient depth vs. N (pbc) for selected dissipative rules.

6 Status and next steps

The graph-level dissipative classification is complete and checkpointed; Tier-1b per-class attractor types are exact and validated. The protected-coherence gap identifies the rules whose full attractor algebra (cross-class noiseless subsystems, weak-symmetry coherences) exceeds the graph's resolution — the targets for a complete Tier-1b/Tier-2 treatment. All numbers derive from `results/*.jsonl` (engine_version 1a.1) and the exact superoperator engine; regenerated by `python -m qca_fragmentation.quantum.dissipative_report`.